Name _____ Period _____ Date _____

6-1 Lesson Quiz: Key Features of Exponential Functions

short form questions keep their Savvas numbers (1, 3, 5); show your work

1. Which function represents the exponential function $f(x) = 3^x$ after a vertical stretch by a factor of 8 and a reflection across the x-axis?

Ⓐ $g(x) = 3^{-8x}$

Ⓑ $g(x) = -3^{8x}$

Ⓒ $g(x) = 8 \cdot 3^{-x}$

Ⓓ $g(x) = -8 \cdot 3^x$

Explain your choice: which part of $g(x)$ makes the reflection, and which part makes the stretch?

_____

_____

3. For the function $f(x) = \left(\frac{1}{2}\right) \cdot 6^x$, identify the domain, range, y-intercept, and asymptote. Write one of the given phrases or values in each blank.

linear	quadratic	exponential	all real numbers	0.5	0	0

Domain: _____ y-intercept: _____

Range: $\{y \mid y >$ _____$\}$ Asymptote: $y =$ _____

Function type: _____

5. Which of the following functions has a greater growth factor than the function shown in the graph?

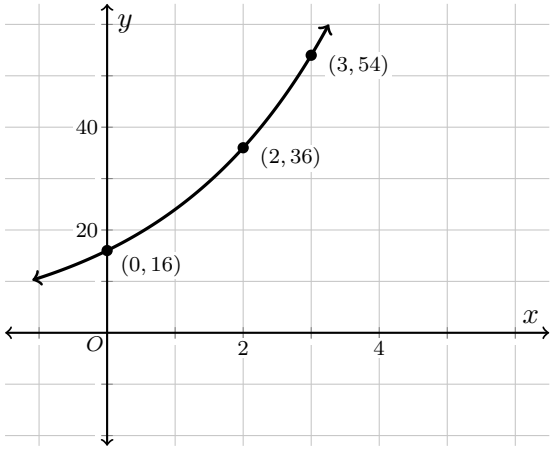
Ⓐ $f(x) = \left(\frac{3}{2}\right) \cdot 3^x$

Ⓑ $f(x) = 8\left(\frac{3}{2}\right)^x$

Ⓒ $f(x) = 20(1.4)^x$

Ⓓ $f(x) = 12(0.8)^x$

Show your work: the growth factor of the graphed function is $b =$ _____



Scoring: Question 1 and Question 5 are 10 points each. Question 3 is a bonus of up to 5 points. Total out of 20.

6-1 Lesson Quiz, short form: answer key

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1. **D, $g(x) = -8 \cdot 3^x$.** The factor 8 multiplies every y-value (vertical stretch) and the minus sign makes each y-value its opposite (reflection across the x-axis): $g(x) = -8f(x)$. A and B put the 8 in the exponent (a horizontal change, not a stretch of the y-values); C has the stretch but its $-x$ reflects across the y-axis. Savvas quiz item 1, DOK 2, Skills Review and Practice F28. Taught by Example 2 ($g(x) = -3^x = -f(x)$ reflects across the x-axis). **10 points:** 6 for D, 4 for the explanation (the $-$ reflects, the 8 stretches).

3. **Domain: all real numbers; Range: $\{y \mid y > 0\}$; y-intercept: 0.5; Asymptote: $y = 0$; Function type: exponential.** $a = \frac{1}{2}$ is the y-intercept because $6^0 = 1$; with no vertical shift the asymptote stays $y = 0$ (Example 2's rule: the asymptote moves only with a shift k). Savvas quiz item 3, DOK 2, F13. Taught by Example 1 (cut); Example 2's intercept and asymptote rules and the packet's definition line carry it. **Bonus 5:** 1 per correct blank.

5. **A, $f(x) = \left(\frac{3}{2}\right) \cdot 3^x$.** The graph's growth factor is $b = \frac{54}{36} = 1.5$ (and $\frac{36}{16} = 2.25 = 1.5^2$ over two steps), so the graphed function is $16(1.5)^x$. Choice A's growth factor is $3 > 1.5$. B has the same factor 1.5, C has $1.4 < 1.5$, and D (0.8) is decay. Savvas quiz item 5, DOK 2, F14. Taught by Example 5 (growth factor as the ratio of y-values one unit apart). **10 points:** 6 for A, 4 for $b = 1.5$ shown from the points. A student who divides $\frac{36}{16}$ and calls it b forgets that x went up by 2.

Items left off this form (available for a retake or Form B): Savvas item 2, 15,000 blue trout with $f(x) = 15{,}000\left(\frac{9}{8}\right)^x$: after 5 years the population increases by about 12,000 (answer D; $f(5) \approx 27{,}030$; DOK 2, F14; it is the cut Examples 3 and 4); Savvas item 4, the rabbit population of Springfield, Ohio, 144,000 in 2016 and decreasing by about 7.2% per year: $P(t) = 144{,}000(0.928)^t$, graphed as a decay curve from $(0, 144{,}000)$ toward the t-axis (DOK 2, F13 and F14; it is the cut Example 3 and its decay-model box). Both are full Savvas quiz only because no kept Example teaches them.

IXL replacement: SmartScore 70 or above on RSN, EFD and 2YT replaces this quiz with 20/20.